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SYSTEMS AND METHODS FOR PERFORMING REPAIRS TO A VEHICLE

Abstract

A system for instructing a user on repairing a vehicle configured to (i) receive, from the user computing device, a request to repair a vehicle, the vehicle including a damaged portion, (ii) prompt the user by displaying a prompt on a user interface of the user computing device to transmit one or more images of the damaged portion of the vehicle; (iii) execute a machine learning model by inputting the one or more images of the damaged portion of the vehicle, the machine learning model configured to determine damage to the vehicle based on at least the one or more images of the vehicle and to determine one or more repair facilities capable of performing repairs to the damage to the vehicle; and (iv) cause the one or more repair facilities to be displayed on the user computing device.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/069,059, filed Dec. 20, 2022, entitled, “SYSTEMS AND METHODS FOR PERFORMING REPAIRS TO A VEHICLE,” which claims priority to U.S. patent application Ser. No. 16/169,547, filed Oct. 24, 2018 and issued as U.S. Pat. No. 11,587,046 on Feb. 21, 2023, entitled, “SYSTEMS AND METHODS FOR PERFORMING REPAIRS TO A VEHICLE,” which claims priority to U.S. Provisional Patent Application No. 62/576,704, filed Oct. 25, 2017, entitled “SYSTEMS AND METHODS FOR PERFORMING REPAIRS TO A VEHICLE,” the entire contents and disclosure of which are all hereby incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

[0002] The present disclosure relates to repairing damage to a vehicle and, more particularly, to a network-based system and method for instructing a user on repairing a vehicle.

BACKGROUND

[0003] In most cases of damage to a vehicle, the damage must be reviewed by an individual, such as an appraiser, to determine a cost to repair the damage. This may be done to inform the owner of the vehicle on how much it will cost to repair, or to inform an insurance provider of the cost to repair. In order to determine the cost to repair the owner of the vehicle must either transport the vehicle to the appraiser for inspection, or the appraiser must travel to the vehicle. Depending on the amount of damage to the vehicle, it may not be worthwhile to require either the owner or the appraiser to travel for the inspection. Furthermore, in some cases the appraisers are repair shops that may or may not have capacity to repair the vehicle at the time of the appraisal. Thus, after getting the appraisal, the owner may have to return with the vehicle or take it elsewhere to get it repaired. There exists a need to automate this process to reduce the need for travel and improve the speed of this process.

BRIEF SUMMARY

[0004] The present embodiments may relate to systems and methods for instructing a user on repairing a vehicle. The system may include a repair assistant (RA) computer system, one or more insurer network computer devices, one or more user devices potentially associated with at least one camera, and/or one or more repair facility computer devices. The RA computer system may be associated with an insurance network or may be merely in communication with an insurance network.

[0005] The RA computer system may be configured to: (i) receive, from the user, a request to repair a vehicle, including information about the vehicle, where the user interface is associated with an application on the user computer device, and where the user logs into the application to make

the request to repair, where the information about the vehicle includes one or more of a make of the vehicle, a model of the vehicle, a year of the vehicle, a location of the damage on the vehicle, identification of the vehicle, identification of the user, and one or more photographs of the vehicle; (ii) in response to the request to repair, transmit, to the user computer device, a link to the user interface, where the link is transmitted in at least one of a short message service (SMS) message, a multimedia messaging server (MMS) message, and an email message; (iii) present, to the user via the user computer device, a user interface to allow the user to search for a repair facility to repair the vehicle; (iv) receive, from the user via the user interface, a search request including a name of a repair facility; (v) request that the user reenter the name of the repair facility if the name is not recognized; (vi) query one or more databases to recognize the repair facility based on the name; (vii) if the name is recognized, retrieve, from the one or more databases, facility information about the repair facility; (viii) determine, via the one or more databases, whether the repair facility is a select service location based on the facility information; (ix) receive, from the user via the user interface, a search location for a repair facility, where the search location includes at least one of an address, a zip code, a municipality, and a present location of the user computer device; (x) query, one or more databases, to determine a plurality of repair facilities within a predetermined distance of the search location; (xi) filter the plurality of repair facilities to generate a subset of repair facilities, where the subset of repair facilities are select service locations; (xii) present, to the user via the user interface, the subset of repair facilities; (xiii) receive, from the user via the user interface, a selection of one of the repair facilities of the subset of repair facilities; (xiv) determine whether the selected repair facility is a select service location, where a select service location is a pre-authorized repair facility; (xv) if the selected repair facility is a select service location, the at least one processor is further programmed to: (a) transfer the information about the vehicle to a computer device associated with the selected repair facility; (b) request that the user upload one or more photographs of at least one of the vehicle and the damage to the vehicle; (c) receive the one or more photographs from the user computer device; and/or (d) transmit the one or more photographs to the selected repair facility; and/or (xvi) if the selected repair facility is not a select service location, the at least one processor is further programmed to: (a) request, from the user via the user interface, drivability information including whether or not the vehicle is able to be safely driven; (b) if the determination is that the vehicle is not able to be safely driven, schedule a phone call to continue the repair process; and/or (c) if the determination is that the vehicle is able to be safely driven, the at least one processor is further programmed to: (1) present, to the user via the user interface, a search interface to search for an inspection site; (2) receive, from the user via the user interface, a selection of an inspection site; (3) present, to the user via the user interface, a calendar to allow the user to set an appointment with the inspection site; and/or (4) determine an appointment based on the user selections in the user interface and appointment information from a computer device associated with the selected inspection site.

[0006] In one aspect, a computer system for instructing a user on repairing a vehicle may be provided. The computer system may include at least one processor (and/or associated transceiver) in communication with at least one memory device. The computer system is in communication with a user computer device associated with the user. The at least one processor (and/or associated transceiver) may be configured or programmed to: (i) receive, from the user, a request to repair a vehicle, including information about the vehicle; (ii) present, to the user via the user computer device, a user interface to allow the user to search for a repair facility to repair the vehicle; (iii) receive, from the user via the user interface, a selection of a repair facility; (iv) determine whether the selected repair facility is a select service location, where a select service location is a pre-authorized repair facility; and (v) if the selected repair facility is a select service location, transfer the information about the vehicle to a computer device associated with the selected repair facility to facilitate quickly and efficiently repairing the vehicle. The computer system may have additional, less, or alternate functionality, including that discussed elsewhere herein.

[0007] In another aspect, a computer-implemented method for instructing a user on repairing a vehicle may be provided. The method may be implemented on a repair assistant ("RA") computer system including at least one processor in communication with at least one memory device. The RA computer system is further in communication with a user computer device associated with the user. The method may include: (i) receiving, from the user, a request to repair a vehicle, including information about the vehicle; (ii) presenting, to the user via the user computer device, a user interface to allow the user to search for a repair facility to repair the vehicle; (iii) receiving, from user via the user interface, a selection of a repair facility; (iv) determining whether the selected repair facility is a select service location, where a select service location is a pre-authorized repair facility; and (v) if the selected repair facility is a select service location, transferring the information about the vehicle to a computer device associated with the selected repair facility to facilitate quickly and efficiently repairing the vehicle. The method may have additional, less, or alternate functionality, including that discussed elsewhere herein.

[0008] In at least one further aspect, at least one non-transitory computer-readable storage media having computer-executable instructions embodied thereon may be provided. When executed by at least one processor, the computer-executable instructions may cause the processor to: (i) receive, from a user, a request to repair a vehicle, including information about the vehicle; (ii) present, to the user via a user computer device, a user interface to allow the user to search for a repair facility to repair the vehicle; (iii) receive, from the user via the user interface, a selection of a repair facility; (iv) determine whether the selected repair facility is a select service location, where a select service location is a pre-authorized repair facility; (v) if the selected repair facility is a select service location, transfer the information about the vehicle to a computer device associated with the selected repair facility; and (vi) if the selected repair facility is not a select service location, the at least one processor is further programmed to: (a) request, from the user via the user interface, drivability information including whether or not the vehicle is able to be safely driven; (b) if the determination is that the vehicle is not able to be safely driven, schedule a phone call to continue the repair process; and (c) if the determination is that the vehicle is able to be safely driven, the at least one processor is further programmed to: (1) present, to the user via the user interface, a search interface to search for an inspection site; and (2) receive, from the user via the user interface, a selection of an inspection site to facilitate quickly and efficiently repairing the vehicle. The computer-executable instructions may have additional, less, or alternate functionality, including that discussed elsewhere herein.

[0009] Advantages will become more apparent to those skilled in the art from the following description of the preferred embodiments which have been shown and described by way of illustration. As will be realized, the present embodiments may be capable of other and different embodiments, and their details are capable of modification in various respects. Accordingly, the drawings and description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The Figures described below depict various aspects of the systems and methods disclosed therein. It should be understood that each Figure depicts an embodiment of a particular aspect of the disclosed systems and methods, and that each of the Figures is intended to accord with a possible embodiment thereof. Further, wherever possible, the following description refers to the reference numerals included in the following Figures, in which features depicted in multiple Figures are designated with consistent reference numerals.

[0011] There are shown in the drawings arrangements which are presently discussed, it being understood, however, that the present embodiments are not limited to the precise arrangements and

are instrumentalities shown, wherein:

[0012] FIG. 1 illustrates a flow chart of an exemplary process of instructing a user in how and where to repair a vehicle, in accordance with the present disclosure.

[0013] FIG. 2 illustrates a flow chart of an exemplary computer-implemented process for one aspect of the process of instructing a user in how and where to repair a vehicle as shown in FIG. 1, in accordance with the present disclosure.

[0014] FIG. 3 illustrates a simplified block diagram of an exemplary computer system for implementing the processes shown in FIGS. 1 and 2.

[0015] FIG. 4 illustrates an exemplary configuration of a user computer device, in accordance with one embodiment of the present disclosure.

[0016] FIG. 5 illustrates an exemplary configuration of a server computer device, in accordance with one embodiment of the present disclosure.

[0017] FIG. 6 illustrates a diagram of components of one or more exemplary computing devices that may be used in the system shown in FIG. 3.

[0018] FIG. 7 illustrates a view of a user interface for allowing the user to search for a repair facility using the system shown in FIG. 3.

[0019] FIG. 8 illustrates a view of a user interface for displaying a map with multiple repair facilities for a user to choose from using the system shown in FIG. 3.

[0020] FIG. 9 illustrates a view of a user interface for displaying a confirmation to a user using the system shown in FIG. 3.

[0021] The Figures depict preferred embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the systems and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION OF THE DRAWINGS

[0022] The present embodiments may relate to, inter alia, systems and methods for instructing a user in how and where to repair a vehicle. In one exemplary embodiment, the process may be performed by a repair assistant (“RA”) computer device. In the exemplary embodiment, the RA computer device may be in communication with a user computer device, such as a mobile computer device, insurer network computer devices, one or more repair facilities, and one or more inspection sites. In the exemplary embodiment, the RA computer device is in communication with a user computer device, where the RA computer device transmits data to the user computer device to be displayed to the user and receives the user's inputs from the user computer device.

[0023] In the exemplary embodiment, a user reports a loss. This loss may be damage to a vehicle or other object, such as due to a vehicular accident. In other examples, the damage may be due to random chance or Mother Nature, such as hail damage or damage from a falling tree limb. In the exemplary embodiment, the user utilizes a user computer device, such as a mobile computer device, to report the loss. In the exemplary embodiment, the user may utilize an application, or website, associated with an insurance provider to report the loss. In other embodiments, the user may call an insurance agent to report the loss. In the exemplary embodiment, the user provides information about the loss, such as, but not limited to, the make, model, and/or year of the vehicle, the circumstances surrounding the damage, location of the damage on the vehicle, identification of the vehicle, identification of the user, and questions about the current condition of the vehicle. These questions may include, but are not limited to, does the hood open and close freely, does the trunk open and close freely, do all of the doors open and close freely, does the vehicle appear to have any suspension damage, and are any fluids leaking. Additionally, the user indicates whether to not he or she plans to repair vehicle. In some further embodiments, the user may also upload photographs and/or images of the vehicle and/or the damage to the vehicle. In some embodiments, RA computer device may request that the user upload additional photographs for being forwarded to the selected repair facility. Systems for uploading photos of damage to a vehicle and for

reviewing those photos to determine the amount of damage to the vehicle based on those photos can be found in U.S. patent application Ser. No. 16/158,137 and U.S. patent application Ser. No. 16/158,118, both filed Oct. 11, 2018, and entitled "SYSTEMS AND METHODS FOR MODEL-BASED ANALYSIS OF DAMAGE TO A VEHICLE." The entire contents and disclosure of both are hereby incorporated by reference herein in their entirety.

[0024] In response to reporting the loss, the user is directed to a specialized user interface. In some embodiments, the user may receive an e-mail or text message (such as a short message service (SMS) message or a multimedia messaging service (MMS) message) that includes a link to the specialized user interface. In other embodiments, the user may have reported the loss in an application associated with the insurance provider, where the user had to provide login credentials to access the application. In these embodiments, the application directs the user to the specialized user interface.

[0025] In the exemplary embodiment, the user is directed to search for a repair facility to repair the vehicle. The user may search by location or by the name of the repair facility. If the user searches by location, then the user may enter one of, but not limited to, an address, a zip code, a municipality, and a present location of the user computer device. The RA computer device searches one or more databases for repair facilities that are within a predetermined distance of the provided location. The predetermined distance may be set by the user, potentially while performing the search. The predetermined distance may also be set by the RA computer device based on the condition of the vehicle and how safe it is to operate the vehicle and/or by the distance that the vehicle may be towed.

[0026] In the exemplary embodiment, the RA computer device filters the display to only display repair facilities that are designated as select service locations. Select service locations are repair facilities that are pre-authorized by the insurance provider to perform repair work. In the exemplary embodiment, these repair facilities have pre-existing relationships with the insurance provider. In some embodiments, the insurance provider has a certification program that it applies to repair facilities to determine whether or not the repair facility qualifies to be a select service location.

[0027] The RA computer device displays the repair facilities. In the exemplary embodiment, the RA computer device only displays repair facilities that are considered select service locations. In the exemplary embodiment, the RA computer device displays the repair facilities on a map. In some further embodiments, the map is centered on the location that the user entered. In some embodiments, the user may move the map displayed to see the repair facilities in different areas. The user may also zoom in and out on the map. In other embodiments, the RA computer device may display the repair locations in a list. As an enhancement, each repair location on the list may include the distance between the location that the user entered and the corresponding repair facility. The information next to each repair location may also include, but is not limited to, shop repair specialty, customer ratings, hours of operation, etc.

[0028] The RA computer device receives the user's selection of a repair facility from the user computer device. In some embodiments, the RA computer device displays information about the selection and requests that the user confirms the selection. If the user does not confirm the selection, the RA computer device displays the repair facilities and allows the user to make a different selection. Once the RA computer device has received the user's confirmed selection of a repair facility, the RA computer device transfers the information about the loss provided by the user to the repair facility. In the exemplary embodiment, the RA computer device transmits the information about the loss and the user to a computer device associated with the repair facility, such as repair facility computer device. In some further embodiments, the RA computer device assists the user in setting up an appointment with the repair facility to repair the vehicle. Then the RA computer device transmits a confirmation to the user to acknowledge that their vehicle is registered to be repaired by the selected repair facility.

[0029] If the user searches by name, then the user may enter part or all of a name of a repair

facility. The RA computer device searches one or more databases for repair facilities that match the entered information. In some situations, there may be more than one repair facility that matches, such as in the case of a chain of repair facilities. In these situations, the RA computer device displays all or a portion of repair facilities that match the entered name. In some embodiments, the RA computer device may only display those repair facilities that are within a predetermined distance, a/k/a less than **50** miles from the user's residential or work address. The user may select the repair facility from a list of similarly named facilities and/or from a single selection. If the user does not see the repair facility that they desired on the list, the user may search repair facilities again, either by name or location. In some embodiments, the RA computer device displays information about the selection and requests that the user confirms the selection. If the user does not confirm the selection, the user may search repair facilities again, either by name or location.

[0030] In the exemplary embodiment, the RA computer device searches one or more databases to determine whether the selected repair facility is a select service location. As described above, select service locations are repair facilities with a pre-existing relationship with the insurance provider, where the insurance provider has certified and/or pre-authorized the repair facility to perform covered repair on vehicles. If the RA computer device determines that the confirmed repair facility is a select service location, the RA computer device transfers the information about the loss provided by the user to the repair facility. In the exemplary embodiment, the RA computer device transmits the information about the loss and the user to the repair facility computer device. In some further embodiments, the RA computer device assists the user in setting up an appointment with the repair facility to repair the vehicle. Then the RA computer device transmits a confirmation to the user to acknowledge that their vehicle is registered to be repaired by the selected repair facility.

[0031] If the RA computer device determines that the confirmed repair facility is not a select service location, the RA computer device determines whether or not the vehicle is drivable. In some embodiments, the RA computer device may ask the user if the vehicle is safe to drive to the selected repair facility. In other embodiments, the RA computer device may determine whether or not the vehicle is safe to drive from the information about the loss that the user provided previously. If the RA computer device determines that the vehicle is not drivable, then the RA computer device sets-up (schedules) a phone call between the user and an insurance agent or other representative of the insurance provider.

[0032] If the RA computer device determines that the vehicle is drivable, the RA computer device instructs the user that an inspection of the damage to the vehicle and the vehicle itself is necessary prior to taking the vehicle to the repair facility for repair. The RA computer device requests a location from the user of a location near where the user desires the inspection to take place. This location may be near the current location of the vehicle, the residence of the user, and/or the workplace of the user. The location may be, but is not limited to, an address, a zip code, a municipality, and a present location of the user computer device.

[0033] The RA computer device receives the entered location. Based on the entered location, the RA computer device searches for inspection sites within a predetermined distance, at which the vehicle may be inspected. In the exemplary embodiment, the inspection sites may include select service location repair facilities and/or other locations where the insurance provider has set-up inspection sites. The RA computer device displays the inspection sites to the user through the user computer device. The RA computer device receives from the user a selection of an inspection site.

[0034] The RA computer device sets an inspection appointment for the vehicle with the user and the inspection site. In the some embodiments, the RA computer device communicates with a computer device associated with the inspection site and retrieves a calendar of appointments, such as inspection site computer device. Then the RA computer device receives a user's selection of a date and time for the appointment and compares the user's selection to the calendar. If there is a conflict, then the RA computer device suggests another time and/or requests that the user select a different time. In other embodiments, the RA computer device transmits the user's selected date and

time to the inspection site computer device, where the inspection site computer device determines whether or not to accept the selected date and time for the appointment. Once the appointment is set, the RA computer device transmits a confirmation of the inspection appointment to the user. [0035] While the above describe the object to be repaired as being a vehicle, the object may be one of any other object that needs to be analyzed to determine the amount of damage that the object has sustained. In some further embodiments, the object may be, but is not limited to, a personal possession, such as an antique clock, a piece of artwork, and/or a piece of furniture.

Exemplary Process for Instructing a User in How and Where to Repair a Vehicle

[0036] FIG. 1 illustrates a flow chart of an exemplary process **100** of instructing a user in how and where to repair a vehicle, in accordance with the present disclosure. In the exemplary embodiment, process **100** is performed by a computer device associated with an insurance provider, such as repair assistant (RA) computer device **310** (shown in FIG. 3). In other embodiments, process **100** is performed by a computer device in communication with an insurance provider. In the exemplary embodiment, RA computer device **310** is in communication with a user computer device, such as a mobile computer device, for example user computer device **305** (shown in FIG. 3). In this embodiment, RA computer device **310** performs process **100** by transmitting data to the user computer device **305** to be displayed to the user and receives the user's inputs from user computer device **305**.

[0037] In the exemplary embodiment, a user reports **105** a loss. This loss may be damage to a vehicle or other object, such as due to a vehicular accident. In other examples, the damage may be due to random chance or Mother Nature, such as hail damage or damage from a falling tree limb. In the exemplary embodiment, the user utilizes a user computer device **305**, such as a mobile computer device, to report **105** the loss. In the exemplary embodiment, the user may utilize an application, or website, associated with an insurance provider to report **105** the loss. In other embodiments, the user may call an insurance agent to report **105** the loss. In the exemplary embodiment, the user provides information about the loss, such as, but is not limited to, the make, model, and/or year of the vehicle, the circumstances surrounding the damage, location of the damage on the vehicle, identification of the vehicle, identification of the user, and questions about the current condition of the vehicle. These questions may include, but are not limited to, does the hood open and close freely, does the trunk open and close freely, do all of the doors open and close freely, does the vehicle appear to have any suspension damage, and are any fluids leaking. Additionally, the user indicates whether to not he or she plans to repair vehicle. In some further embodiments, the user may also upload photographs and/or images of the vehicle and/or the damage to the vehicle. In some embodiments, RA computer device may request that the user upload additional photographs for being forwarded to the selected repair facility.

[0038] In response to reporting **105** the loss, the user is directed to a specialized user interface. In some embodiments, the user may receive an e-mail or text message (such as a SMS message or a MMS message) that includes a link to the specialized user interface. In other embodiments, the user may have reported **105** the loss in an application associated with the insurance provider, where the user had to provide login credentials to access the application. In these embodiments, the application directs the user to the specialized user interface.

[0039] In the exemplary embodiment, the user is directed to search **110** for a repair facility to repair the vehicle. The user may search **110** by location or by the name of the repair facility. If the user searches **110** by location, then the user may enter one of, but not limited to, an address, a zip code, a municipality, and a present location of the user computer device **305**. The RA computer device **310** searches one or more databases for repair facilities that are within a predetermined distance of the provided location. The predetermined distance may be set by the user, potentially while performing the search. The predetermined distance may also be set by the RA computer device **310** based on the condition of the vehicle and how safe it is to operate the vehicle and/or by the distance that the vehicle may be towed.

[0040] In the exemplary embodiment, the RA computer device **310** filters the display to only display repair facilities that are designated as select service locations. Select service locations are repair facilities that are pre-authorized by the insurance provider to perform repair work. In the exemplary embodiment, these repair facilities have pre-existing relationships with the insurance provider. In some embodiments, the insurance provider has a certification program that it applies to repair facilities to determine whether or not the repair facility qualifies to be a select service location.

[0041] The RA computer device **310** displays **115** the repair facilities. In the exemplary embodiment, the RA computer device **310** only displays **115** repair facilities that are considered select service locations. In the exemplary embodiment, the RA computer device **310** displays **115** the repair facilities on a map. In some further embodiments, the map is centered on the location that the user entered. In some embodiments, the user may move the map displayed to see the repair facilities in different areas. The user may also zoom in and out on the map. In other embodiments, the RA computer device **310** may display **115** the repair locations in a list. As an enhancement, each repair location on the list may include the distance between the location that the user entered and the corresponding repair facility. The information next to each repair location may also include, but is not limited to, shop repair specialty, customer ratings, hours of operation, etc.

[0042] The RA computer device **310** receives **120** the user's selection of a repair facility from the user computer device **305**. In some embodiments, the RA computer device **310** displays information about the selection and requests that the user confirms the selection. If the user does not confirm the selection, the RA computer device **310** displays **115** the repair facilities and allows the user to make a different selection. Once the RA computer device **310** has received **120** the user's confirmed selection of a repair facility, the RA computer device **310** transfers **125** the information about the loss provided by the user to the repair facility. In the exemplary embodiment, the RA computer device **310** transmits the information about the loss and the user to a computer device associated with the repair facility, such as repair facility computer device **330** (shown in FIG. 3). In some further embodiments, the RA computer device **310** assists the user in setting up an appointment with the repair facility to repair the vehicle. Then the RA computer device **310** transmits **130** a confirmation to the user to acknowledge that their vehicle is registered to be repaired by the selected repair facility.

[0043] If the user searches **110** by name, then the user may enter part or all of a name of a repair facility. The RA computer device **310** searches one or more databases for repair facilities that match the entered information. In some situations, there may be more than one repair facility that matches, such as in the case of a chain of repair facilities. In these situations, the RA computer device **310** displays all or a portion of repair facilities that match the entered name. In some embodiments, the RA computer device **310** may only display those repair facilities that are within a predetermined distance, a/k/a less than 50 miles from the user's residential or work address. The user may select the repair facility from a list of similarly named facilities and/or from a single selection. If the user does not see the repair facility that they desired on the list, the user may search **110** repair facilities again, either by name or location. In some embodiments, the RA computer device **310** displays information about the selection and requests that the user confirms the selection. If the user does not confirm the selection, the user may search **110** repair facilities again, either by name or location.

[0044] In the exemplary embodiment, the RA computer device **310** searches one or more databases to determine **135** whether the selected repair facility is a select service location. As described above, select service locations are repair facilities with a pre-existing relationship with the insurance provider, where the insurance provider has certified and/or pre-authorized the repair facility to perform covered repair on vehicles. If the RA computer device **310** determines **135** that the confirmed repair facility is a select service location, the RA computer device **310** transfers **125** the information about the loss provided by the user to the repair facility. In the exemplary

embodiment, the RA computer device **310** transmits the information about the loss and the user to the repair facility computer device **330**. In some further embodiments, the RA computer device **310** assists the user in setting up an appointment with the repair facility to repair the vehicle. Then the RA computer device **310** transmits **130** a confirmation to the user to acknowledge that their vehicle is registered to be repaired by the selected repair facility.

[0045] If the RA computer device **310** determines **135** that the confirmed repair facility is not a select service location, the RA computer device **310** determines **140** whether or not the vehicle is drivable. In some embodiments, the RA computer device **310** may ask the user if the vehicle is safe to drive to the selected repair facility. In other embodiments, the RA computer device **310** may determine **140** whether or not the vehicle is safe to drive from the information about the loss that the user provided previously. If the RA computer device **310** determines **140** that the vehicle is not drivable, then the RA computer device **310** sets-up a phone call between the user and an insurance agent or other representative of the insurance provider.

[0046] If the RA computer device **310** determines **140** that the vehicle is drivable, the RA computer device **310** instructs the user that an inspection of the damage to the vehicle and the vehicle itself is necessary prior to taking the vehicle to the repair facility for repair. The RA computer device **310** requests a location from the user of a location near where the user desires the inspection to take place. This location may be near the current location of the vehicle, the residence of the user, and/or the workplace of the user. The location may be, but is not limited to, an address, a zip code, a municipality, and a present location of the user computer device **305**.

[0047] The RA computer device **310** receives **150** the entered location. Based on the entered location, the RA computer device **310** searches for inspection sites within a predetermined distance, at which the vehicle may be inspected. In the exemplary embodiment, the inspection sites may include select service location repair facilities and/or other locations where the insurance provider has set-up inspection sites. The RA computer device **310** displays **155** the inspection sites to the user through the user computer device **305**. The RA computer device **310** receives **160** from the user a selection of an inspection site.

[0048] The RA computer device **310** sets **165** an inspection appointment for the vehicle with the user and the inspection site. In the some embodiments, the RA computer device **310** communicates with a computer device associated with the inspection site and retrieves a calendar of appointments, such as inspection site computer device **335** (shown in FIG. 3). Then the RA computer device **310** receives a user's selection of a date and time for the appointment and compares the user's selection to the calendar. If there is a conflict, then the RA computer device **310** suggests another time and/or requests that the user select a different time. In other embodiments, the RA computer device **310** transmits the user's selected date and time to the inspection site computer device **335**, where the inspection site computer device **335** determines whether or not to accept the selected date and time for the appointment. Once the appointment is set **165**, the RA computer device **310** transmits **170** a confirmation of the inspection appointment to the user.

[0049] While the above describe the object to be repaired as being a vehicle, the object may be one of any other object that needs to be analyzed to determine the amount of damage that the object has sustained. In some further embodiments, the object may be, but is not limited to, a personal possession, such as an antique clock, a piece of artwork, and/or

Exemplary Computer-Implemented Method for Instructing a User in How and Where to Repair a Vehicle

[0050] FIG. 2 illustrates a flow chart of an exemplary computer-implemented process **200** for one aspect of the process **100** of instructing a user in how and where to repair a vehicle as shown in FIG. 1, in accordance with the present disclosure. Process **200** may be implemented by a computing device, for example damage analysis ("RA") computer device **310** (shown in FIG. 3). In the exemplary embodiment, RA computer device **310** may be in communication with a user computer device **305** (shown in FIG. 3), such as a mobile computer device, an insurer network

computer device **325** (shown in FIG. **3**), one or more repair facility computer devices **330** (shown in FIG. **3**), and one or more inspection site computer devices **335** (shown in FIG. **3**).

[0051] In the exemplary embodiment, RA computer device **310** may receive **205**, from the user, a request to repair a vehicle, including information about the vehicle. In the exemplary embodiment, the information about the vehicle includes, but is not limited to, the make, model, and/or year of the vehicle, the circumstances surrounding the damage, location of the damage on the vehicle, identification of the vehicle, identification of the user, and questions about the current condition of the vehicle. These questions may include, but are not limited to, does the hood open and close freely, does the trunk open and close freely, do all of the doors open and close freely, does the vehicle appear to have any suspension damage, and are any fluids leaking. Additionally, the user indicates whether to not he or she plans to repair vehicle. In some further embodiments, the user may also upload photographs and/or images of the vehicle and/or the damage to the vehicle. In some embodiments, RA computer device may request that the user upload additional photographs for being forwarded to the selected repair facility.

[0052] In response to the request, RA computer device **310** may present **210**, to the user via the user computer device, a user interface to allow the user to search for a repair facility to repair the vehicle. In some embodiments, RA computer device **310** may transmit a link to a user interface, such as in a SMS message, a MMS message, or an email to the user. In other embodiments, the user may login to an application associated with the insurance provider to request the repair. The application may direct the user to the user interface. In the exemplary embodiment, RA computer device **310** transmits information about the user interface to the user computer device **305** and the user computer device **305** displays the user interface to the user. In this embodiment, the user computer device **305** receives inputs from the user and transmits those inputs to RA computer device **310**.

[0053] In the exemplary embodiment, the user interface allows the user to search for a repair facility based on the name or the location of the facility. RA computer device **310** may receive, from the user via the user interface, a search request including a name of a repair facility. RA computer device **310** may request that the user reenter the name of the repair facility if the name is not recognized. RA computer device **310** may then query one or more databases to recognize the repair facility based on the name. If the name is recognized, RA computer device **310** may retrieve, from the one or more databases, facility information about the repair facility.

[0054] RA computer device **310** may receive, from the user via the user interface, a search location for a repair facility, where the search location may be at least one of an address, a zip code, a municipality, and a present location of the user computer device. RA computer device **310** may then query, one or more databases, to determine a plurality of repair facilities within a predetermined distance of the search location. In the exemplary embodiment, RA computer device **310** filters the plurality of repair facilities to generate a subset of repair facilities, where the subset of repair facilities are select service locations. Select service locations are repair facilities that are pre-authorized by the insurance provider to perform repair work. In the exemplary embodiment, these repair facilities have pre-existing relationships with the insurance provider. In some embodiments, the insurance provider has a certification program that it applies to repair facilities to determine whether or not the repair facility qualifies to be a select service location. RA computer device **310** then presents, to the user via the user interface, the subset of repair facilities.

[0055] In the exemplary embodiment, RA computer device **310** receives **215**, from the user via the user interface, a selection of a repair facility. RA computer device **310** may receive **215**, from the user via the user interface, a selection of one of the repair facilities of the subset of repair facilities. Based on the selection, RA computer device **310** determines **220** whether the selected repair facility is a select service location. RA computer device **310** may determine **220**, via the one or more databases, whether the repair facility is a select service location based on the facility information provided by one or more databases. If the repair facility was chosen based on the location search,

and only select service locations were displayed, RA computer device **310** automatically determines **220** that the repair facility is a select service location.

[0056] If the selected repair facility is a select service location, RA computer device **310** transfers **225** the information about the vehicle to a computer device associated with the selected repair facility, such as repair facility computer device **330**. In some embodiments, RA computer device **310** transmits a confirmation to the user. RA computer device **310** may also assist the user in setting up an appointment to repair the vehicle.

[0057] If the selected repair facility is not a select service location, RA computer device **310** may request, from the user via the user interface, drivability information including whether or not the vehicle is able to be safely driven. If RA computer device **310** determines that the vehicle is not able to be safely driven, RA computer device **310** may schedule a phone call to continue the repair process. In some embodiments, RA computer device **310** may initiate call to an insurance provider agent or representative. In other embodiments, RA computer device **310** may transmit a request for communication to an insurer network computer device **325**, wherein the insurer network computer device **325** may initiate a call between the user and a representative of the insurance provider and/or schedule a call in the future. For example, RA computer device **310** may ask the user to enter when the user wishes to be called.

[0058] In some embodiments, RA computer device **310** may communicate with one or more insurer network computer devices **325** to setup the call. If RA computer device **310** determines that the vehicle is able to be safely driven, RA computer device **310** may present, to the user via the user interface, a search interface to search for an inspection site. The computer system may also receive, from the user via the user interface, a selection of an inspection site. In some embodiments, RA computer device **310** may present, to the user via the user interface, a calendar to allow the user to set an appointment with the inspection site. RA computer device **310** may also determine an appointment based on the user selections in the user interface and appointment information from a computer device associated with the selected inspection site, such as inspection site computer device **335**.

[0059] In some embodiments, RA computer device **310** transmits views for the user interface to the user computer device **305**. The user computer device **305** is configured to receive and display the views. RA computer device **310** receives user inputs received by the user computer device **305** from the user and transmitted to RA computer device **310**.

[0060] While the above describe the object to be repaired as being a vehicle, the object may be one of any other object that needs to be analyzed to determine the amount of damage that the object has sustained. In some further embodiments, the object may be, but is not limited to, a personal possession, such as an antique clock, a piece of artwork, and/or a piece of furniture.

Exemplary Computer Network

[0061] FIG. **3** depicts a simplified block diagram of an exemplary computer system **300** for implementing process **100** shown in FIG. **1** and process **200** shown in FIG. **2**. In the exemplary embodiment, computer system **300** may be used for instructing a user in how and where to repair a vehicle. As described below in more detail, a repair assistant (“RA”) computer device **310** may be configured to (i) receive, from a user, a request to repair a vehicle, including information about the vehicle; (ii) present, to the user via user computer device **305**, a user interface to allow the user to search for a repair facility to repair the vehicle; (iii) receive, from the user via the user interface, a selection of a repair facility; (iv) determine whether the selected repair facility is a select service location, where a select service location is a pre-authorized repair facility; and (v) if the selected repair facility is a select service location, transfer the information about the vehicle to a computer device associated with the selected repair facility.

[0062] In the exemplary embodiment, user computer devices **305** are computers that include a web browser or a software application, which enables user computer devices **305** to access remote computer devices, such as RA computer device **310** and insurer network computer devices **325**,

using the Internet or other network. More specifically, user computer devices **305** may be communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem. User computer devices **305** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. In some embodiments, user computer device **305** is in communication with a camera. In some of these embodiments, the camera is integrated into user computer device **305**. In other embodiments, the camera is a separate device that is in communication with user computer device **305**, such as through a wired connection, i.e. a universal serial bus (USB) connection.

[0063] A database server **315** may be communicatively coupled to a database **320** that stores data. In one embodiment, database **320** may include vehicle information, select service locations, inspection sites, and repair facility information. In the exemplary embodiment, database **320** may be stored remotely from RA computer device **310**. In some embodiments, database **320** may be decentralized. In the exemplary embodiment, the user may access database **320** via user computer device **305** by logging onto RA computer device **310**, as described herein.

[0064] RA computer device **310** may be communicatively coupled with one or more user computer devices **305**. In some embodiments, RA computer device **310** may be associated with, or is part of a computer network associated with an insurance provider, or in communication with insurer network computer devices **325**. In other embodiments, RA computer device **310** may be associated with a third party and is merely in communication with the insurer network computer devices **325**. More specifically, RA computer device **310** is communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem.

[0065] RA computer device **310** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. In the exemplary embodiment, RA computer device **310** hosts an application or website that allows the user to access the functionality described herein. In some further embodiments, user computer device **305** includes an application that facilitates communication with RA computer device **310**.

[0066] In the exemplary embodiment, insurer network computer devices **325** include one or more computer devices associated with an insurance provider. In the exemplary embodiment, insurance provider is associated with the user and the user has an insurance policy that insures the object with insurance provider. In the exemplary embodiment, insurer network computer devices **325** include a web browser or a software application, which enables insurer network computer devices **325** to access remote computer devices, such as RA computer device **310** and database server **315**, using the Internet or other network. More specifically, insurer network computer devices **325** may be communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem. Insurer network computer devices **325** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. In some embodiments, insurer network computer devices **325** may access database

320 to select service locations, inspection sites, and/or repair facility information.

[0067] In the exemplary embodiment, repair facility computer devices **330** include computer devices associated with repair facilities capable of repairing object. In the exemplary embodiment, repair facility computer devices **330** include a web browser or a software application, which enables repair facility computer devices **330** to access remote computer devices, such as RA computer device **310**, using the Internet or other network. More specifically, repair facility computer devices **330** may be communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem. Repair facility computer devices **330** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. In some embodiments, repair facility computer devices **330** may communicate with RA computer device **310** to schedule repair appointments. In some embodiments, repair facilities also function as inspection sites. In these embodiments, repair facility computer devices **330** may communicate with RA computer device **310** to schedule inspection appointments. Repair facility computer devices **330** may communicate with database **320** to retrieve information about the vehicle to be repaired.

[0068] In the exemplary embodiment, inspection site computer devices **335** include computer devices associated with inspection sites capable of inspecting the damage to the object. In the exemplary embodiment, inspection site computer devices **335** include a web browser or a software application, which enables inspection site computer devices **335** to access remote computer devices, such as RA computer device **310**, using the Internet or other network. More specifically, inspection site computer devices **335** may be communicatively coupled to the Internet through many interfaces including, but not limited to, at least one of a network, such as the Internet, a local area network (LAN), a wide area network (WAN), or an integrated services digital network (ISDN), a dial-up-connection, a digital subscriber line (DSL), a cellular phone connection, and a cable modem. Inspection site computer devices **335** may be any device capable of accessing the Internet including, but not limited to, a desktop computer, a laptop computer, a personal digital assistant (PDA), a cellular phone, a smartphone, a tablet, a phablet, wearable electronics, smart watch, or other web-based connectable equipment or mobile devices. In some embodiments, inspection site computer devices **335** may communicate with RA computer device **310** to schedule inspection appointments. Inspection site computer devices **335** may communicate with database **320** to retrieve information about the vehicle to be inspected.

Exemplary Client Device

[0069] FIG. 4 depicts an exemplary configuration **400** of user computer device **402**, in accordance with one embodiment of the present disclosure. In the exemplary embodiment, user computer device **402** may be similar to, or the same as, user computer device **305** (shown in FIG. 3). User computer device **402** may be operated by a user **401**. User computer device **402** may include, but is not limited to, user computer devices **305**, insurer network computer devices **325**, repair facility computer devices **330**, and inspection site computer devices **335** (all shown in FIG. 3). User computer device **402** may include a processor **405** for executing instructions. In some embodiments, executable instructions may be stored in a memory area **410**. Processor **405** may include one or more processing units (e.g., in a multi-core configuration). Memory area **410** may be any device allowing information such as executable instructions and/or transaction data to be stored and retrieved. Memory area **410** may include one or more computer readable media.

[0070] User computer device **402** may also include at least one media output component **415** for presenting information to user **401**. Media output component **415** may be any component capable of conveying information to user **401**. In some embodiments, media output component **415** may

include an output adapter (not shown) such as a video adapter and/or an audio adapter. An output adapter may be operatively coupled to processor **405** and operatively coupleable to an output device such as a display device (e.g., a cathode ray tube (CRT), liquid crystal display (LCD), light emitting diode (LED) display, or “electronic ink” display) or an audio output device (e.g., a speaker or headphones).

[0071] In some embodiments, media output component **415** may be configured to present a graphical user interface (e.g., a web browser and/or a client application) to user **401**. A graphical user interface may include, for example, an interface for viewing repair facility locations. In some embodiments, user computer device **402** may include an input device **420** for receiving input from user **401**. User **401** may use input device **420** to, without limitation, select a repair facility and/or inspection site.

[0072] Input device **420** may include, for example, a keyboard, a pointing device, a mouse, a stylus, a touch sensitive panel (e.g., a touch pad or a touch screen), a gyroscope, an accelerometer, a position detector, a biometric input device, and/or an audio input device. A single component such as a touch screen may function as both an output device of media output component **415** and input device **420**.

[0073] User computer device **402** may also include a communication interface **425**, communicatively coupled to a remote device such as RA computer device **310** (shown in FIG. 3). Communication interface **425** may include, for example, a wired or wireless network adapter and/or a wireless data transceiver for use with a mobile telecommunications network.

[0074] Stored in memory area **410** are, for example, computer readable instructions for providing a user interface to user **401** via media output component **415** and, optionally, receiving and processing input from input device **420**. A user interface may include, among other possibilities, a web browser and/or a client application. Web browsers enable users, such as user **401**, to display and interact with media and other information typically embedded on a web page or a website from RA computer device **310**. A client application may allow user **401** to interact with, for example, RA computer device **310**. For example, instructions may be stored by a cloud service, and the output of the execution of the instructions sent to the media output component **415**.

Exemplary Server Device

[0075] FIG. 5 depicts an exemplary configuration **500** of a server computer device **501**, in accordance with one embodiment of the present disclosure. In the exemplary embodiment, server computer device **501** may be similar to, or the same as, RA computer device **310** (shown in FIG. 3). Server computer device **501** may include, but is not limited to, RA computer device **310**, insurer network computer devices **325**, repair facility computer device **330**, inspection site computer device **335**, and database server **315** (all shown in FIG. 3). Server computer device **501** may also include a processor **505** for executing instructions. Instructions may be stored in a memory area **510**. Processor **505** may include one or more processing units (e.g., in a multi-core configuration).

[0076] Processor **505** may be operatively coupled to a communication interface **515** such that server computer device **501** is capable of communicating with a remote device such as another server computer device **501**, RA computer device **310**, insurer network computer devices **325**, repair facility computer device **330**, inspection site computer device **335**, and user computer devices **305** (shown in FIG. 3) (for example, using wireless communication or data transmission over one or more radio links or digital communication channels). For example, communication interface **515** may receive requests from user computer devices **305** via the Internet, as illustrated in FIG. 3.

[0077] Processor **505** may also be operatively coupled to a storage device **534**. Storage device **534** may be any computer-operated hardware suitable for storing and/or retrieving data, such as, but not limited to, data associated with database **320** (shown in FIG. 3). In some embodiments, storage device **534** may be integrated in server computer device **501**. For example, server computer device **501** may include one or more hard disk drives as storage device **534**.

[0078] In other embodiments, storage device **534** may be external to server computer device **501** and may be accessed by a plurality of server computer devices **501**. For example, storage device **534** may include a storage area network (SAN), a network attached storage (NAS) system, and/or multiple storage units such as hard disks and/or solid state disks in a redundant array of inexpensive disks (RAID) configuration.

[0079] In some embodiments, processor **505** may be operatively coupled to storage device **534** via a storage interface **520**. Storage interface **520** may be any component capable of providing processor **505** with access to storage device **534**. Storage interface **520** may include, for example, an Advanced Technology Attachment (ATA) adapter, a Serial ATA (SATA) adapter, a Small Computer System Interface (SCSI) adapter, a RAID controller, a SAN adapter, a network adapter, and/or any component providing processor **505** with access to storage device **534**.

[0080] Processor **505** may execute computer-executable instructions for implementing aspects of the disclosure. In some embodiments, the processor **505** may be transformed into a special purpose microprocessor by executing computer-executable instructions or by otherwise being programmed. For example, the processor **505** may be programmed with the instruction such as illustrated in FIGS. **1** and **2**.

Exemplary Computer Device

[0081] FIG. **6** depicts a diagram **600** of components of one or more exemplary computing devices **610** that may be used in system **300** shown in FIG. **3**. In some embodiments, computing device **610** may be similar to RA computer device **310**. Database **620** may be coupled with several separate components within computing device **610**, which perform specific tasks. In this embodiment, database **620** may include vehicle information **622**, select service locations **624**, inspection sites **626**, and repair facility information **628**. In some embodiments, database **620** is similar to database **320** (shown in FIG. **3**).

[0082] Computing device **610** may include the database **620**, as well as data storage devices **630**. Computing device **610** may also include a communication component **640** for receiving **205** a request to repair a vehicle, receiving **215** a selection of a repair facility, and transferring **225** the information about the vehicle (all shown in FIG. **2**). Computing device **610** may further include a presenting component **650** for presenting **210** a user interface (shown in FIG. **2**). Moreover, computing device **610** may include a determining component **660** for determining **220** whether the selected repair facility is a select service location (shown in FIG. **2**). A processing component **670** may assist with execution of computer-executable instructions associated with the system.

Exemplary User Interface

[0083] FIG. **7** illustrates a view **700** of a user interface for allowing the user to search for a repair facility using system **300** (shown in FIG. **3**). In the exemplary embodiment, view **700** of user interface is displayed on user computer device **305** (shown in FIG. **3**).

[0084] View **700** displays a search bar for allowing a user to search for a repair facility, such as described in Step **110** of process **100** (shown in FIG. **1**). As described above, the user may enter a shop name, address, city, state, and/or zip code to search for a repair facility. View **700** is similar to the view of the search bar for searching for an inspection site, as described in Step **150** of process **100**.

Exemplary User Interface

[0085] FIG. **8** illustrates a view **800** of a user interface for displaying a map with multiple repair facilities for a user to choose from using system **300** (shown in FIG. **3**). In the exemplary embodiment, view **800** of user interface is displayed on user computer device **305** (shown in FIG. **3**).

[0086] View **800** displays a map including a plurality of repair facilities for the user to select. In view **800**, the user has clicked on the first repair facility. The view **800** displays information about the first repair facility and allows the user to select this repair facility by pressing the button labeled "Use Shop." If the user selects a different repair facility, the view **800** will change and display

information about that repair facility. This map with selections is also described in Step **115** of process **100** (shown in FIG. **1**). View **800** is similar to the view of the map for selecting an inspection site for inspecting the vehicle, as described in Step **155** of process **100**. In the exemplary embodiment, entering a location in view **700** causes the user interface to proceed to view **800** to display repair facilities and/or inspection sites within a predetermined distance of the location to be displayed.

Exemplary User Interface

[0087] FIG. **9** illustrates a view **900** of a user interface for displaying a confirmation to a user using system **300** (shown in FIG. **3**). In the exemplary embodiment, view **900** of user interface is displayed on user computer device **305** (shown in FIG. **3**).

[0088] View **900** displays a confirmation to the user to acknowledge that their vehicle is registered to be repaired by the selected repair facility. View **900** displays information to the user to allow them to proceed to the selected repair facility, as described in Step **130** of process **100** (shown in FIG. **1**). View **900** is similar to the view of the confirmation for setting an appointment for an inspection of the vehicle, as described in Step **170** of process **100**.

Exemplary Embodiments & Functionality

[0089] In one aspect, a computer system for instructing a user in how and where to repair a vehicle may be provided. The computer system may include at least one processor in communication with at least one memory device. The computer system may be in communication with a user computer device associated with the user. The at least one processor may be configured or programmed to: (1) receive, from the user, a request to repair a vehicle, including information about the vehicle; (2) present, to the user via the user computer device, a user interface to allow the user to search for a repair facility to repair the vehicle; (3) receive, from the user via the user interface, a selection of a repair facility; (4) determine whether the selected repair facility is a select service location, wherein a select service location is a pre-authorized repair facility; and (5) if the selected repair facility is a select service location, transfer the information about the vehicle to a computer device associated with the selected repair facility.

[0090] If the selected repair facility is not a select service location, a further enhancement may be where the computer system may request, from the user via the user interface, drivability information including whether or not the vehicle is able to be safely driven. If the determination is that the vehicle is not able to be safely driven, the computer system may schedule a phone call to continue the repair process. In some embodiments, the computer system may initiate call to an insurance provider agent or representative. In other embodiments, the computer system may transmit a request for communication to an insurer network computer device, wherein the insurer network computer device may initiate a call between the user and a representative of the insurance provider and/or schedule a call in the future. For example, the computer system may ask the user to enter when the user wishes to be called. If the determination is that the vehicle is able to be safely driven, the computer system may present, to the user via the user interface, a search interface to search for an inspection site. The computer system may also receive, from the user via the user interface, a selection of an inspection site.

[0091] A further enhancement may be where the computer system may present, to the user via the user interface, a calendar to allow the user to set an appointment with the inspection site. The computer system may also determine an appointment based on the user selections in the user interface and appointment information from a computer device associated with the selected inspection site.

[0092] The computer system may achieve the above results by receiving, from the user via the user interface, a search request including a name of a repair facility. The computer system may request that the user reenter the name of the repair facility if the name is not recognized. The computer system may then query one or more databases to recognize the repair facility based on the name. If the name is recognized, the computer system retrieves, from the one or more databases, facility

information about the repair facility. The computer system may then determine, via the one or more databases, whether the repair facility is a select service location based on the facility information. [0093] The computer system may achieve the above results by receiving, from the user via the user interface, a search location for a repair facility, where the search location may be at least one of an address, a zip code, a municipality, and a present location of the user computer device. The computer system may then query, one or more databases, to determine a plurality of repair facilities within a predetermined distance of the search location. The computer system may filter the plurality of repair facilities to generate a subset of repair facilities, wherein the subset of repair facilities are select service locations. The computer system may also present, to the user via the user interface, the subset of repair facilities and receive, from the user via the user interface, a selection of one of the repair facilities of the subset of repair facilities.

[0094] A further enhancement may be where in response to the request to repair the vehicle, the computer system may transmit, to the user computer device, a link to the user interface, wherein the link is transmitted in at least one of a SMS message, an MMS message, and an email message.

[0095] A further enhancement may be where the user interface is associated with an application on the user computer device. In this enhancement the user logs into the application to make the request to repair the vehicle.

[0096] The computer system may achieve the above results by including the user interface. The computer system may then transmit views for the user interface to the user computer device, where the user computer device is configured to receive and display the views. The computer system may then receive user inputs received by the user computer device from the user and transmitted to the computer system.

[0097] A further enhancement may be where the information about the vehicle includes one or more of a make of the vehicle, a model of the vehicle, a year of the vehicle, a location of the damage on the vehicle, identification of the vehicle, and identification of the user.

Machine Learning & Other Matters

[0098] The computer-implemented methods discussed herein may include additional, less, or alternate actions, including those discussed elsewhere herein. The methods may be implemented via one or more local or remote processors, transceivers, and/or sensors (such as processors, transceivers, and/or sensors mounted on vehicles or mobile devices, or associated with smart infrastructure or remote servers), and/or via computer-executable instructions stored on non-transitory computer-readable media or medium.

[0099] Additionally, the computer systems discussed herein may include additional, less, or alternate functionality, including that discussed elsewhere herein. The computer systems discussed herein may include or be implemented via computer-executable instructions stored on non-transitory computer-readable media or medium.

[0100] A processor or a processing element may be trained using supervised or unsupervised machine learning, and the machine learning program may employ a neural network, which may be a convolutional neural network, a deep learning neural network, a reinforced or reinforcement learning module or program, or a combined learning module or program that learns in two or more fields or areas of interest. Machine learning may involve identifying and recognizing patterns in existing data in order to facilitate making predictions for subsequent data. Models may be created based upon example inputs in order to make valid and reliable predictions for novel inputs.

[0101] Additionally or alternatively, the machine learning programs may be trained by inputting sample (e.g., training) data sets or certain data into the programs, such as image data of objects to be analyzed (e.g., vehicle damage), mobile device data, and/or vehicle telematics data. The machine learning programs may utilize deep learning algorithms that may be primarily focused on pattern recognition, and may be trained after processing multiple examples. The machine learning programs may include Bayesian program learning (BPL), voice recognition and synthesis, image or object recognition, optical character recognition, and/or natural language processing-either

individually or in combination. The machine learning programs may also include natural language processing, semantic analysis, automatic reasoning, and/or other types of machine learning, such as deep learning, reinforced learning, or combined learning.

[0102] Supervised and unsupervised machine learning techniques may be used. In supervised machine learning, a processing element may be provided with example inputs and their associated outputs, and may seek to discover a general rule that maps inputs to outputs, so that when subsequent novel inputs are provided the processing element may, based upon the discovered rule, accurately predict the correct output. In unsupervised machine learning, the processing element may be required to find its own structure in unlabeled example inputs. The unsupervised machine learning techniques may include clustering techniques, cluster analysis, anomaly detection techniques, multivariate data analysis, probability techniques, unsupervised quantum learning techniques, associate mining or associate rule mining techniques, and/or the use of neural networks. In some embodiments, semi-supervised learning techniques may be employed. In one embodiment, machine learning techniques may be used to extract data about the object, vehicle, user, damage, needed repairs, costs and/or incident from vehicle data, insurance policies, geolocation data, image data, and/or other data.

[0103] In the exemplary embodiment, a processing element may be trained by providing it with a large sample of repair and vehicle damage data with known characteristics or features. Such information may include, for example, information associated with a shape, size, and/or type of damage as well as the qualifications of various repair facilities and past choices of users related to different repair facilities based on information such as insurance policies, geolocation data, image data, and/or other data.

[0104] Based upon these analyses, the processing element may learn how to identify characteristics and patterns that may then be applied to analyzing repair data. For example, the processing element may learn, with the user's permission or affirmative consent, to identify the repair facilities most appropriate for certain types of vehicles and certain types of damages. The processing element may also learn how to identify attributes of different repair facilities that make a user more or less likely to choose that repair facility. This information may be used to determine which repair facilities should be made or qualified as select service locations.

Technical Advantages

[0105] The aspects described herein may be implemented as part of one or more computer components such as a client device and/or one or more back-end components, such as a repair assessment engine, for example. Furthermore, the aspects described herein may be implemented as part of a computer network architecture and/or a cognitive computing architecture that facilitates communications between various other devices and/or components. Thus, the aspects described herein address and solve issues of a technical nature that are necessarily rooted in computer technology.

[0106] For instance, aspects include analyzing various sources of data to determine damage to an object, such as a vehicle, determine a repair facility to repair the damage, and assist a user in setting up an appointment. In doing so, the aspects overcome issues associated with the inconvenience of manually driving to an inspection facility and requiring multiple inspections prior to repair of the damage. Furthermore, these aspects reduce the number of trips required to travel to a repair facility that may require one trip for the inspection and another trip for the actual repair. Without the improvements suggested herein, additional processing and memory usage would be required to perform such coordination. Additional technical advantages include, but are not limited to: i) improved speed and responsiveness in repairing an object; ii) aligning the owner of the object to be repaired with a repair facility capable of performing the repair; iii) aligning the owner of the object to be repaired with a repair facility that is covered by the insurance provider to perform the repair; iv) efficient transfer of data to the repair facility about the damage prior to arrival at the facility; v) analyzing policies to determine the most efficient repair to the object; and vi) providing an efficient

user interface to receive the required data in an efficient and accurate manner. Additional technical advantages are described in other sections of the specification.

[0107] Furthermore, the embodiments described herein improve upon existing technologies, and improve the functionality of computers, by more accurately predicting or identifying which repair facilities that a user may select. The present embodiments improve the speed, efficiency, and accuracy in which such calculations and processor analysis may be performed. Due to these improvements, the aspects address computer-related issues regarding efficiency over conventional techniques. Thus, the aspects also address computer related issues that are related to efficiency metrics and ease of use, for example.

Additional Considerations

[0108] As will be appreciated based upon the foregoing specification, the above-described embodiments of the disclosure may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof. Any such resulting program, having computer-readable code means, may be embodied or provided within one or more computer-readable media, thereby making a computer program product, i.e., an article of manufacture, according to the discussed embodiments of the disclosure. The computer-readable media may be, for example, but is not limited to, a fixed (hard) drive, diskette, optical disk, magnetic tape, semiconductor memory such as read-only memory (ROM), and/or any transmitting/receiving medium, such as the Internet or other communication network or link. The article of manufacture containing the computer code may be made and/or used by executing the code directly from one medium, by copying the code from one medium to another medium, or by transmitting the code over a network.

[0109] These computer programs (also known as programs, software, software applications, “apps”, or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” “computer-readable medium” refers to any computer program product, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The “machine-readable medium” and “computer-readable medium,” however, do not include transitory signals. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

[0110] As used herein, a processor may include any programmable system including systems using micro-controllers, reduced instruction set circuits (RISC), application specific integrated circuits (ASICs), logic circuits, and any other circuit or processor capable of executing the functions described herein. The above examples are example only, and are thus not intended to limit in any way the definition and/or meaning of the term “processor.”

[0111] As used herein, the terms “software” and “firmware” are interchangeable, and include any computer program stored in memory for execution by a processor, including RAM memory, ROM memory, EPROM memory, EEPROM memory, and non-volatile RAM (NVRAM) memory. The above memory types are example only, and are thus not limiting as to the types of memory usable for storage of a computer program.

[0112] In one embodiment, a computer program is provided, and the program is embodied on a computer readable medium. In an exemplary embodiment, the system is executed on a single computer system, without requiring a connection to a sever computer. In a further embodiment, the system is being run in a Windows® environment (Windows is a registered trademark of Microsoft Corporation, Redmond, Washington). In yet another embodiment, the system is run on a mainframe environment and a UNIX® server environment (UNIX is a registered trademark of X/Open Company Limited located in Reading, Berkshire, United Kingdom). In a further embodiment, the

system is run on an iOS® environment (iOS is a registered trademark of Cisco Systems, Inc. located in San Jose, CA). In yet a further embodiment, the system is run on a Mac OS® environment (Mac OS is a registered trademark of Apple Inc. located in Cupertino, CA). In still yet a further embodiment, the system is run on Android® OS (Android is a registered trademark of Google, Inc. of Mountain View, CA). In another embodiment, the system is run on Linux® OS (Linux is a registered trademark of Linus Torvalds of Boston, MA). The application is flexible and designed to run in various different environments without compromising any major functionality. [0113] In some embodiments, the system includes multiple components distributed among a plurality of computing devices. One or more components may be in the form of computer-executable instructions embodied in a computer-readable medium. The systems and processes are not limited to the specific embodiments described herein. In addition, components of each system and each process can be practiced independent and separate from other components and processes described herein. Each component and process can also be used in combination with other assembly packages and processes.

[0114] As used herein, an element or step recited in the singular and preceded by the word “a” or “an” should be understood as not excluding plural elements or steps, unless such exclusion is explicitly recited. Furthermore, references to “example embodiment” or “one embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

[0115] The patent claims at the end of this document are not intended to be construed under 35 U.S.C. § 112 (f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being expressly recited in the claim(s).

[0116] This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A computer system for identifying repairs for a vehicle using a machine learning model, the computer system comprising at least one processor in communication with at least one memory device, the computer system in communication with a user computing device associated with the user, the at least one processor is programmed to: receive, from the user computing device, a request to repair a vehicle, the vehicle including a damaged portion; prompt the user by displaying a prompt on a user interface of the user computing device to transmit one or more images of the damaged portion of the vehicle; execute the machine learning model by inputting the one or more images of the damaged portion of the vehicle, the machine learning model configured to determine damage to the vehicle based on the one or more images of the vehicle and determine one or more repair facilities capable of performing repairs to the damage to the vehicle; and cause the one or more repair facilities to be displayed on the user computing device.

2. The computer system of claim 1, wherein the at least one processor is further programmed to: generate, by the at least one processor, a plurality of questions about current conditions of the vehicle; transmit, from the at least one processor to the user computer device, a plurality of instructions to cause the user interface of the user computer device to display the plurality of questions about the current conditions of the vehicle; and receive, by the at least one processor from the user computer device via the user interface of the user computer device, a plurality of answers about the current conditions of the vehicle in response to the plurality of questions.

3. The computer system of claim 1, wherein the at least one processor is further programmed to receive an indication from the user of whether the user plans to repair the vehicle.
4. The computer system of claim 3, wherein the at least one processor is further programmed to: request, by the at least one processor from the user computer device via the user interface, one or more additional images of the vehicle based on an indication of whether the user plans to repair the vehicle, wherein the request includes instructions for taking the one or more images of the vehicle; and receive, by the at least one processor from the user computer device via the user interface, the one or more additional images of the vehicle captured by a camera of the user computer device.
5. The computer system of claim 1, wherein the at least one processor is further programmed to transmit, from the at least one processor to the user computer device, an electronic message including instructions to display a link to a user interface to be displayed on the user computer device, wherein, when the link is selected, the link causes additional instructions to be transmitted to the user computer device to cause display of the user interface on the user computer device.
6. The computer system of claim 1, wherein the at least one processor is further programmed to: retrieve, by the at least one processor, a calendar of appointments from an inspection facility computer device associated with a selected inspection facility; receive, by the at least one processor via the user interface, a user selection of a date and time for an appointment; determine a conflict between the user selection of the date and time for the appointment and the calendar of appointments of the selected inspection facility; and transmit an alternative date and time for an appointment to the user computing device based on the calendar of appointments.
7. The computer system of claim 1, wherein the at least one processor is further programmed to: receive, from the user computer device via the user interface, a search request including a name of a repair facility; query one or more databases to recognize the repair facility based on the name; if the name is recognized, retrieve, from one or more databases, facility information about the repair facility; determine, via the one or more databases, whether the repair facility is a select service location based on the facility information; and request that the user reenter the name of the repair facility if the name is not recognized.
8. The computer system of claim 1, wherein the at least one processor is further programmed to: receive, from the user computer device via the user interface, a search location for a repair facility, wherein the search location includes at least one of an address, a zip code, a municipality, or a present location of the user computer device; wherein the machine learning model is further configured to determine a safe threshold distance based on a condition of the vehicle; query, one or more databases, to determine a plurality of repair facilities within the safe threshold distance of the search location; and present, to the user computer device via the user interface, the plurality of repair facilities.
9. The computer system of claim 1, wherein in response to the request to repair the vehicle, the at least one processor is further programmed to generate a link to the user interface, wherein the link is transmitted to the user computer device in at least one of a short message service (SMS) message, a multimedia messaging server (MMS) message, and an email message.
10. The computer system of claim 1, wherein the at least one processor is further programmed to receive, from the user computing device, information about the vehicle, wherein the information about the vehicle includes one or more of a make of the vehicle, a model of the vehicle, a year of the vehicle, a location of damage on the vehicle, identification of the vehicle, identification of the user, or the one or more images of the vehicle.
11. The computer system of claim 1, wherein the at least one processor is further programmed to train the machine learning model to determine damage to the vehicle and to determine one or more repair facilities capable of performing repairs on the vehicle based on a first subset of vehicle damage data.
12. The computer system of claim 11, wherein the at least one processor is further programmed to train the machine learning model to determine damage to the vehicle and to determine one or more

repair facilities capable of performing repairs on the vehicle based on a second subset of vehicle repair data.

13. The computer system of claim 1, wherein the at least one processor is further programmed to: execute the machine learning model to analyze a plurality of answers from a plurality of vehicles and a plurality of selected repair facilities from a plurality of users; and determine one or more repair facilities to qualify as select service locations based on the executed machine learning model.

14. The computer system of claim 13, wherein the at least one processor is further programmed to determine one or more attributes that make a user more or less likely to choose a repair facility based on the executed machine learning model.

15. A computer-implemented method for identifying repairs for a vehicle using a machine learning model, the method implemented on a repair assistant (“RA”) computer system comprising at least one processor in communication with at least one memory device, the RA computer system further in communication with a user computing device associated with the user, the method comprising: receiving, from the user computing device, a request to repair a vehicle, the vehicle including a damaged portion; prompt the user by displaying a prompt on a user interface of the user computing device to transmit one or more images of the damaged portion of the vehicle; executing the machine learning model by inputting the one or more images of the damage portion of the vehicle, the machine learning model configured to determine damage to the vehicle based on the one or more images of the vehicle and determine one or more repair facilities capable of performing repairs to the damage to the vehicle; and cause the one or more repair facilities to be displayed on the user computing device.

16. The method of claim 15 further comprising: generating, by the at least one processor, a plurality of questions about current conditions of the vehicle; transmitting, from the at least one processor to the user computer device, a plurality of instructions to cause the user interface of the user computer device to display the plurality of questions about the current conditions of the vehicle; and receiving, by the at least one processor from the user computer device via the user interface of the user computer device, a plurality of answers about the current conditions of the vehicle in response to the plurality of questions.

17. The method of claim 16 further comprising transmitting, from the at least one processor to the user computer device, an electronic message including instructions to display a link to a user interface to be displayed on the user computer device, wherein, when the link is selected, the link causes additional instructions to be transmitted to the user computer device to cause display of the user interface on the user computer device.

18. The method of claim 16 further comprising: requesting, by the at least one processor from the user computer device via the user interface, the one or more images of the vehicle based on the plurality of answers, wherein the request includes instructions for taking the one or more images of the vehicle; and receiving, by the at least one processor from the user computer device via the user interface, the one or more images of the vehicle captured by a camera of the user computer device.

19. The method of claim 15 further comprising: retrieving, by the at least one processor, a calendar of appointments from an inspection facility computer device associated with a selected inspection facility; receiving, by the at least one processor via the user interface, a user selection of a date and time for an appointment; determine a conflict between the user selection of the date and time for the appointment and the calendar of appointments of the selected inspection facility; and transmit an alternative date and time for an appointment to the user computing device based on the calendar of appointments.

20. The method of claim 15 further comprising: receiving, from the user computer device via the user interface, a search request including a name of a repair facility; querying one or more databases to recognize the repair facility based on the name; if the name is recognized, retrieving, from the one or more databases, facility information about the repair facility; determining, via the one or more databases, whether the repair facility is a select service location based on the facility

information; and requesting that the user reenter the name of the repair facility if the name is not recognized.

21. At least one non-transitory computer-readable storage media having computer-executable instructions embodied thereon, wherein when executed by at least one processor in communication with at least one memory device and a user computer device associated with a user, the computer-executable instructions cause the processor to: receive, from the user computing device, a request to repair a vehicle, the vehicle including a damaged portion; prompt the user by displaying a prompt on a user interface of the user computing device to transmit one or more images of the damaged portion of the vehicle; execute a machine learning model by inputting the one or more images of the damaged portion of the vehicle, the machine learning model configured to determine damage to the vehicle based on the one or more images of the vehicle and determine one or more repair facilities capable of performing repairs to the damage to the vehicle; and cause the one or more repair facilities to be displayed on the user computing device.
